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Inventor(s)	Liu; Yunxiang et al.

Integrally shielded cable connector

Abstract

A cable assembly with a shielded termination portion where signal conductors of a plurality of cables are directly mounted to tail portions of signal terminals of a cable connector. Shielding in the termination portion may be provided by interaction of a wave-shaped shield, tail portions of ground terminals between tail portions of the signal terminals, a cross bar connecting the ground terminals and an exposed portion of a cable shield. The wave-shaped shield may be mechanically connected to at least the tail portions of the ground terminals and may be configured to press the cable shields into the bar. Such a configuration improves signal transmission performance by providing a consistent shielding at the connection areas where the cables are mounted to the tail portions of the conductive terminals and at the same time eliminates the need of an intermediate circuit board and therefore reduces manufacturing costs.

Inventors:	Liu; Yunxiang (Chengdu, CN), Zeng; Jianke (Chengdu, CN)
Applicant:	Amphenol Commercial Products (Chengdu) Co., Ltd. (Chengdu, CN)
Family ID:	81290141
Assignee:	Amphenol Commercial Products (Chengdu) Co., Ltd. (Chengdu, CN)
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Primary Examiner: Chambers; Travis S

Attorney, Agent or Firm: Wolf, Greenfield & Sacks, P.C.

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/505,437, now U.S. Pat. No. 12,057,663, filed on Oct. 19, 2021, entitled “INTEGRALLY SHIELDED CABLE CONNECTOR,” which claims priority to and the benefit of Chinese Patent Application Serial No. 202120784796.X, filed on Apr. 16, 2021. U.S. patent application Ser. No. 17/505,437, now U.S. Pat. No. 12,057,663, also claims priority to and the benefit of Chinese Patent

Application Serial No. 202110412091.X, filed on Apr. 16, 2021. U.S. patent application Ser. No. 17/505,437, now U.S. Pat. No. 12,057,663, also claims priority to and the benefit of Chinese Patent Application Serial No. 202022373960.6, filed on Oct. 22, 2020. U.S. patent application Ser. No. 17/505,437, now U.S. Pat. No. 12,057,663, also claims priority to and the benefit of Chinese Patent Application Serial No. 202011136063.1, filed on Oct. 22, 2020. The contents of these applications are incorporated herein by reference in their entirety.

FIELD

(1) This application relates generally to electrical interconnection systems, such as those including cables, used to interconnect electronic components.

BACKGROUND

(2) Cables may be used to interconnect electronic components that are separated by a distance. As electronic systems become increasingly more complex, an electrical connector may be used to establish the electrical connections between a plurality of cables and a circuit board on which some of the electronic components to be interconnected are mounted.

BRIEF SUMMARY

(3) Aspects of the present disclosure relate to integrally shielded cable connectors.

(4) Some embodiments relate to a cable connector. The cable connector may include an insulative housing; a plurality of conductive terminals held in a row by the insulative housing, the plurality of conductive terminals comprising pairs of signal terminals separated by ground terminals, each conductive terminal comprising a contact portion, a tail portion opposite the contact portion, and an intermediate portion extending between the contact portion and the tail portion; and a shield mechanism. The shield mechanism may include the ground terminals, a crossbar connecting the tail portions of the ground terminals, and a wave-shaped shield member comprising alternating peak portions and valley portions, wherein each valley portion is mounted to a respective ground terminal, and each peak portion spans a respective pair of signal terminals.

(5) In some embodiments, for each pair of signal terminals, the tail portions of the pair of signal terminals may be disposed in a U-shaped space defined by the crossbar and the tail portions of two adjacent ground terminals.

(6) In some embodiments, each peak portion of the wave-shaped shield member may include an opening sized and positioned to expose at least a portion of the tail portions of a respective pair of signal terminals.

(7) In some embodiments, each valley portion of the wave-shaped shield member may be welded to a respective ground terminal.

(8) In some embodiments, the insulative housing may include a socket configured to receive at least a portion of a circuit board.

(9) In some embodiments, the contact portions of the conductive terminals may curve into the socket of the insulative housing.

(10) Some embodiments relate to a cable assembly. The cable assembly may include a cable connector and a plurality of cables. The cable connector may include a plurality of conductive terminals held in a row, the plurality of conductive terminals comprising pairs of signal terminals separated by ground terminals, each conductive terminal comprising a contact portion, a tail portion opposite the contact portion, and an intermediate portion extending between the contact portion and the tail portion; and a shield mechanism comprising the ground terminals and a crossbar connecting the tail portions of the ground terminals. Each cable of the plurality of cables may include a pair of signal conductors and a shield conductor surrounding the pair of signal conductors. The pair of signal conductors of each cable may be mounted on the tail portions of a respective pair of signal terminals, and the shield conductor of each cable may be electrically connected to the ground terminals through contacting the crossbar.

(11) In some embodiments, the shield mechanism may include a wave-shaped shield member. The

wave-shaped shield member, the ground terminals, the crossbar and the shield conductor together may form shielding around connection areas where the signal conductors of the cables are mounted on the tail portions of the signal terminals.

(12) In some embodiments, the wave-shaped shield may include peak portions and valley portions disposed in alternative. Each valley portion may be attached to a respective ground terminal. Each peak portion may span a respective pair of signal terminals and define a chamber in which a respective pair of signal conductors of a cable may be received. Each peak portion may press against a respective shield conductor of a cable so as to urge the cable towards the crossbar.

(13) In some embodiments, each peak portion may include an opening into the chamber such that at least a portion of a respective connection area may be visible through the opening.

(14) In some embodiments, each peak portion may include a first holding mechanism for increasing retention of the respective shield conductor of the cable.

(15) In some embodiments, the first holding mechanism may include a protrusion or barb protruding from the wave-shaped shield into the chamber.

(16) In some embodiments, the crossbar may include a second holding mechanism for increasing retention of the respective shield conductor of the cable.

(17) In some embodiments, the second holding mechanism may include a protrusion or barb protruding from the crossbar into the chamber.

(18) In some embodiments, the tail portions of the ground terminals, the tail portions of the plurality of pairs of signal terminals and the crossbar may be substantially in a same plane.

(19) Some embodiments relate to a cable connector. The cable connector may include a plurality of conductive terminals, each conductive terminal comprising a contact portion, a tail portion opposite the contact portion, and an intermediate portion extending between the contact portion and the tail portion; a terminal retention mechanism holding the plurality of conductive terminals in two opposed rows, the conductive terminals in each row are aligned therein and comprise pairs of signal terminals separated by ground terminals, the tail portions of the plurality of ground terminals extending beyond the tail portions of the plurality of pairs of signal terminals; and a shield mechanism comprising the ground terminals, a crossbar connecting the tail portions of the ground terminals, a first wave-shaped shield member electrically and mechanically attached to the ground terminals of a first row of conductive terminals, a second wave-shaped shield member mounted to the ground terminals of a second row of conductive terminals.

(20) In some embodiments, the cable connector further comprising a third shield member extending between the two rows of conductive terminals in a plane parallel to the two rows of conductive terminals. The third shield member may be electrically connected to the first and second wave-shaped shield members.

(21) In some embodiments, the cable connector further comprising a third shield member extending between the two rows of conductive terminals in a plane parallel to the two rows of conductive terminals. The third shield member may be electrically insulated from the first and second wave-shaped shield members.

(22) In some embodiments, the third shield member may be embedded in the terminal retention mechanism.

(23) In some embodiments, the terminal retention mechanism may include a retention portion extending vertically and a boss portion extending laterally from the retention portion.

(24) The retention portion may be disposed around the plurality of conductive terminals to retain the plurality of conductive terminals in the two terminal rows, with tail portions of the conductive terminals in each row extending from the retention portion and resting on a corresponding side of the boss portion.

(25) Some embodiments relate to a cable connector for providing an electrical connection between a circuit board and a cable. The cable connector may include an insulative housing; a plurality of conductive terminals held in the insulative housing, each conductive terminal comprising a contact

portion configured for electrically connecting with a conductive portion of the circuit board and a tail portion configured for electrically connecting with an end portion of the cable; an insulative terminal retention mechanism disposed in the insulative housing and holding the plurality of conductive terminals; and a shield mechanism disposed in the insulative housing and configured for holding the end portion in position relative to the tail portion and providing shielding for a connection area where the tail portion is connected with the end portion when the tail portion is electrically connected with the end portion.

(26) In some embodiments, the plurality of conductive terminals may include a signal terminal and ground terminals. The tail portions of the ground terminals may be connected together through a crossbar. The shield mechanism may include a first shield member. The cable may include a signal conductor and a shield conductor surrounding the signal conductor. The cable may be exposed at the end portion in such a manner that the signal conductor extends beyond the shield conductor. The signal conductor may be connected to the tail portion of the signal terminal. The first shield member may be mounted across the signal conductor and the signal terminal onto and in contact with the ground terminals and hold the shield conductor on the crossbar such that the shield conductor is in contact with the crossbar, whereby the first shield member, the ground terminals, the crossbar and the shield conductor together form a shield around a connection area where the signal conductor is connected with the signal terminal.

(27) In some embodiments, the terminal retention mechanism may be configured for retaining the plurality of conductive terminals in two terminal rows mutually opposed and spaced apart. The conductive terminals in each terminal row may be aligned therein and comprise a plurality of pairs of signal terminals and a plurality of ground terminals. The tail portions of the plurality of ground terminals may extend beyond the tail portions of the plurality of pairs of signal terminals and may be connected together through the crossbar. A pair of signal terminals may be arranged between two adjacent ground terminals, with the tail portions of the pair of signal terminals disposed in a U-shaped space defined by the tail portions of the two adjacent ground terminals and the crossbar.

(28) In some embodiments, the cable may include a plurality of cables, each cable comprising a pair of signal conductors and a shield conductor surrounding the pair of signal conductors. The cable may be exposed at the end portion in such a manner that the pair of signal conductors extend beyond the shield conductor. Each signal conductor of the pair of signal conductors may be connected to the tail portion of a corresponding signal terminal of a corresponding pair of signal terminals of the plurality of pairs of signal terminals, and the first shield member may hold the shield conductor on the crossbar such that the shield conductor is in contact with the crossbar.

(29) In some embodiments, the first shield member may be in the form of a pair of first shield plates each mounted from outside of the two terminal rows onto and in contact with a plurality of ground terminals in a corresponding terminal row, extending longitudinally at least along the entire length of the corresponding terminal row and laterally at least along the tail portions of the plurality of ground terminals and the crossbar, and pressing against and being in contact with the shield conductor.

(30) In some embodiments, each of the pair of first shield plates may be a wave-shaped plate having a peak portion and a valley portion. Each valley portion may be mounted onto and in contact with a corresponding ground terminal of the plurality of ground terminals.

(31) Each peak portion may span the signal conductor and the signal terminal and define a chamber in which the end portion of the cable is received. The peak portion may press against and may be in contact with the shield conductor.

(32) In some embodiments, each peak portion may be formed with an aperture open to the chamber to allow at least a connection area where the pair of signal conductors is connected to the tail portions of the corresponding pair of signal terminals to be accessible via the aperture.

(33) In some embodiments, a first portion of each peak portion in contact with the shield conductor may be formed with a first holding mechanism for improving a holding force on the shield

conductor.

(34) In some embodiments, a second portion of the crossbar in contact with the shield conductor may be formed with a second holding mechanism for improving a holding force on the shield conductor.

(35) In some embodiments, the first holding mechanism may include a protrusion or barb protruding from the first portion into the chamber.

(36) In some embodiments, the second holding mechanism may include a protrusion or barb protruding from the second portion into the chamber.

(37) In some embodiments, the tail portions of the plurality of ground terminals, the tail portions of the plurality of pairs of signal terminals and the crossbar may be substantially in the same plane.

(38) In some embodiments, the shield mechanism may include a second shield member disposed between the two terminal rows in the terminal retention mechanism and extending longitudinally at least along the entire length of the two terminal rows and laterally at least along the tail portions of the plurality of ground terminals and the crossbar.

(39) In some embodiments, the second shield member may be connected with the crossbar and/or the pair of first shield plates, or the second shield member is insulated from the crossbar and the pair of first shield plates.

(40) In some embodiments, the second shield member may be in the form of a shield plate.

(41) In some embodiments, the terminal retention mechanism may be of an elongated T-shaped body configuration and include a vertically extending retention portion and a boss portion extending laterally from the retention portion. The retention portion may be disposed around the plurality of conductive terminals to retain the plurality of conductive terminals in the two terminal rows, with tail portions of the conductive terminals in each terminal row extending from the retention portion and resting on a corresponding side of a first side and a second side opposite to the first side of the boss portion.

(42) In some embodiments, the retention portion may include a first positioning protrusion protruding onto the first side and the second side. The first positioning protrusion may be configured for cooperating with the peak portion to enable the pair of first shield plates to be positioned accurately relative to the plurality of ground terminals and preventing the pair of first shield plates from moving in a direction along the first side and the second side relative to the plurality of ground terminals when the pair of first shield plates is mounted onto the plurality of ground terminals.

(43) In some embodiments, the insulative housing may include a first housing portion and a second housing portion configured to be mounted to the first housing portion. The retention portion may be received in a terminal cavity defined by the first housing portion. The boss portion may project from the first housing portion and may be received in the second housing portion. The second housing portion may hold the pair of first shield plates in position when mounted to the first housing portion.

(44) In some embodiments, the contact portions of the conductive terminals in each terminal row may extend cantilevered from the retention portion opposite to the tail portions and into the terminal cavity and together with the contact portions of the conductive terminals in the other terminal row define therebetween a receiving space for receiving the conductive portion of the circuit board. At least one socket may be formed in the first housing portion and open to the terminal cavity to allow for the insertion of the conductive portion of the circuit board into the receiving space between and in contact with the contact portions of the conductive terminals in the two terminal rows.

(45) In some embodiments, the retention portion may include a first attachment structure and the first housing portion may include a second attachment structure. The first attachment structure and the second attachment structure may be configured for cooperating with each other to secure the retention portion in the terminal cavity.

- (46) In some embodiments, the first attachment structure may include a barb protruding from the retention portion and the second attachment structure may include a notch formed in the first housing portion for receiving the barb.
- (47) In some embodiments, the retention portion may include a first positioning structure and the first housing portion may include a second positioning structure. The first positioning structure and the second positioning structure may be configured for cooperating with each other to enable the retention portion to be accurately positioned in the terminal cavity when the retention portion is placed into the terminal cavity.
- (48) In some embodiments, the first positioning structure may include first positioning slots recessed into the retention portion from two longitudinally opposite end faces of the retention portion, respectively, and the second positioning structure may include first positioning ribs protruding into the terminal cavity from inner walls of the terminal cavity.
- (49) In some embodiments, the first housing portion may include a first mounting structure and the second housing portion may include a second mounting structure, the first mounting structure and the second mounting structure may be configured for cooperating with each other to secure the first housing portion and the second housing portion together.
- (50) In some embodiments, the first mounting structure may include two T-shaped protrusions mutually opposed formed across the terminal cavity on the first housing portion and the second mounting structure may include two T-shaped recesses recessed into the second housing portion from two longitudinally opposite end faces of the second housing portion. Each of the two T-shaped protrusions may snap into a corresponding one of the two T-shaped recesses.
- (51) In some embodiments, the boss portion may include a third positioning structure and the first housing portion may include a fourth positioning structure. The third positioning structure and the fourth positioning structure may be configured for cooperating with each other to enable the retention portion to be accurately positioned in the terminal cavity when the retention portion is placed into the terminal cavity.
- (52) In some embodiments, the third positioning structure may include a pair of third positioning ribs protruding from the boss portion from two longitudinally opposite end faces of the boss portion, respectively. The fourth positioning structure may include a second positioning rib formed on the two T-shaped protrusions. The second positioning rib may be received between the pair of third positioning ribs.
- (53) In some embodiments, the second housing portion may be mounted to the first housing portion by a dual injection molding.
- (54) In some embodiments, each valley portion may be mounted to a corresponding ground terminal of the plurality of ground terminals by laser welding.
- (55) Some embodiments relate to an electrical connection system. The electrical connection may include cables and the aforementioned cable connector. The cables may be connected to a plurality of conductive terminals of the cable connector.
- (56) These techniques may be used alone or in any suitable combination. The foregoing summary is provided by way of illustration and is not intended to be limiting.
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Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The following accompanying drawings of the present disclosure are used here as a part of the present disclosure for understanding the present disclosure. The accompanying drawings are not intended to be drawn to scale. For purposes of clarity, not every component may be labeled in every drawing. In drawings:

(2) FIG. 1A is a partially exploded perspective view of a cable connector according to some

embodiments;

- (3) FIG. 1B is a perspective view of the cable connector of FIG. 1A in an assembled state;
- (4) FIG. 1C is a front view of the cable connector of FIG. 1B;
- (5) FIG. 1D is a rear view of the cable connector of FIG. 1B;
- (6) FIG. 2A is a perspective view of a cable of the cable connector of FIG. 1A;
- (7) FIG. 2B is a cross-sectional view of the cable of FIG. 2A;
- (8) FIG. 3A is a perspective view of a row of conductive terminals of the cable connector of FIG. 1A;
- (9) FIG. 3B is a top view of the row of the conductive terminals of FIG. 3A;
- (10) FIG. 4 is a perspective view of a terminal retention mechanism of the cable connector of FIG. 1A;
- (11) FIG. 5 is a perspective view of the cable connector of FIG. 1A with an insulative housing and a shield mechanism removed;
- (12) FIG. 6 is a perspective view of a shield mechanism of the cable connector of FIG. 1A;
- (13) FIG. 7A is a perspective view of a first housing portion of the insulative housing of the cable connector of FIG. 1A; and
- (14) FIG. 7B is a perspective view of a second housing portion of the insulative housing of the cable connector of FIG. 1A.

LIST OF REFERENCE NUMERALS

(15) **1** cable connector **100** insulative housing **100a** first housing portion **100b** second housing portion **101** terminal cavity **103** first socket **113** second attachment structure **115** second positioning structure **117** first mounting structure **118** second mounting structure **119** fourth positioning structure **121a**, **121b** end faces **200** conductive terminals **200a**, **200b** terminal rows **201** ground terminal **203a** first signal terminal **203b** second signal terminal **205** contact portion **207** tail portion **209** body portion **211** crossbar **213** U-shaped space **300** terminal retention mechanism **301** retention portion **303** boss portion **305a** first side **305b** second side **307** first positioning protrusion **311** slot **313** first attachment structure **315** first positioning structure **316a**, **316b** end faces **319** third positioning structure **320a**, **320b** end faces **321** second positioning slot **400** shield mechanism **401** first shield member **401a**, **401b** a pair of first shield plates **403** peak portion **405** valley portion **407** chamber **409** aperture **411** first retention mechanism **413** second shield member **500** cable **501a** first signal conductor **501b** second signal conductor **503a** first insulator **503b** second insulator **505** shield conductor **507** external insulator **600** solder **S** receiving space

DETAILED DESCRIPTION

(16) The inventors have recognized and appreciated designs for cable connectors that may improve signal transmission performance and reduce manufacturing cost at the same time. The cable connector may be configured to support signal conductors of cables being directly mounted to tail portions of the signal terminals, which eliminates the need of an intermediate circuit board. The cable connector may also include an integrally formed shield mechanism, which provides consistent shielding at the connection areas where the cables are mounted to the tail portions of the conductive terminals and therefore improves signal transmission performance. The shield mechanism may also aid in terminating cables to the cable connector by facilitating mechanical connection of the cable to the terminals of the connector as well as forming electrical connections between cable shields and conductive terminals serving as grounds within the connector.

(17) In some embodiments, the cable connectors may have terminals configured with contact portions configured to mate with a conductive pad in a mating structure. That mating structure may be, for example, a mating connector with blade-like terminals exposed in an insulative tongue, with the exposed portions of the terminals serving as a mating pad. In other embodiments, the pads may be pads along an edge of a printed circuit board containing electronic components to be connected to other parts of an electronic system through the cables terminated to the cable connector. In this configuration, a cable connector may be made without an intermediate circuit board as was used in

conventional cable connectors to connect cables to conductive terminals of the connector. Cable connectors as described herein, in comparison to such conventional cable connectors, have lower manufacturing cost and avoid inconsistent shielding, which lead to poor signal transmission performance in conventional cable connectors especially, for example, at high frequency.

(18) In some embodiments, the integrally formed shield mechanism may include a row of ground terminals connected by a crossbar at their tail portions. In some embodiments, the integrally formed shield mechanism may include one or more wave-shaped shield members that may each include alternating peak portions and valley portions. The valley portions may be mounted to respective ground terminals. The peak portions may span respective pairs of signal terminals separated by the ground terminals. The peak portions may press against the shield conductors of respective cables and press the shield conductors of the respective cables to the crossbar. The peak portions may include holding mechanisms for enhancing the forces applied to the shield conductors of the respective cables.

(19) The peak portions may include openings sized and positioned to expose at least a portion of the tail portions of respective pairs of signal terminals such that, from the openings, any possible shorting between the pair of signal terminals caused by, for example, soldering the signal conductors of the cables to the signal terminals may be observed and resolved. In some embodiments, the integrally formed shield mechanism may include a shield member extending in a plane between two rows of conductive terminals. The shield member may be embedded in a portion of connector housing. In some embodiments, the shield member is electrically connected to the wave-shaped shield members. In some alternative embodiments, the shield member is not electrically connected to the wave-shaped shield members, even if mechanically connected.

(20) Embodiments of a cable connector are described in detail below in conjunction with the accompanying drawings. It should be appreciated by the skilled person in the art that these embodiments are not meant to form any limitation on the present application.

(21) FIGS. 1A to 1D illustrate a cable connector **1** according to an embodiment of the present application. As shown in FIGS. 1A to 1D, the cable connector **1** comprises an insulative housing **100**, a plurality of conductive terminals (which may also be referred to as “conductive tabs”) **200** held in the insulative housing **100**, an insulative terminal retention mechanism (which may also be referred to as a “plastic retainer”) **300** disposed in the insulative housing **100** and retaining the plurality of conductive terminals **200**, and a shield mechanism **400** disposed in the insulative housing **100**. Terminal retention mechanism **300** is an insulative body, which may be formed in one or more pieces, configured to hold the terminals.

(22) In the illustrated embodiment, cable connector **1** is configured for providing an electrical connection between a mating structure, such as a circuit board (not shown), and a cable **500**. The circuit board, for example, may be a paddle card in a mating connector or may be a daughter card in an electronic system. The circuit board may have an edge with a width fitting within socket **103** or may have a portion of an edge configured to fit within socket **103**. Such a circuit board may be formed with a tab sized to fit within socket **103** extending from the edge and/or recesses perpendicular to the edge sized to receive walls of housing portion **100a** bounding socket **103** such that portions of the board between the recesses may fit within socket **103**. Regardless of the configuration of the circuit board, it may have pads against which contacts of the conductive terminals **200** may press to make a separable connection to the mating component.

(23) Each conductive terminal **200** includes a tail portion configured for electrically connecting with an end portion of the cable **500** and a contact portion configured for electrically connecting with a conductive portion of a mating component, such as a pad on the circuit board, as will be described in detail below in conjunction with FIGS. 2A-2B, 3A-3B and 5. The shield mechanism **400** is configured for holding the end portion of the cable in position relative to the tail portions and providing shielding for a connection area where the tail portions are connected with the end portion when the tail portion is electrically connected with the end portion, as will be described in

detail below in conjunction with FIGS. 1A and 6.

(24) FIGS. 2A and 2B schematically illustrate a non-limiting example of the cable **500** suitable for connecting to a circuit board through the cable connector **1** shown in FIG. 1A. As shown in FIGS. 2A and 2B, each cable **500** includes one or more signal conductors and a shield conductor surrounding the signal conductors. In the illustrated embodiment, there are two signal conductors, forming a drainless, twinax cable. In order to establish an electrical connection with the conductive terminals **200** of the cable connector **1** or other electronic components, the cable **500** may be processed at the end portion in such a manner that the shield conductor is exposed, and the signal conductors extend beyond the shield conductor and insulative material surrounding the signal conductors within the bulk cable, as will be described in detail below.

(25) Each cable **500** includes a central pair of signal conductors (which may also be referred to as “inner core wires”) extending parallel to each other, hereinafter referred to as the first signal conductor **501a** and the second signal conductor **501b**. Each of the first signal conductor **501a** and the second signal conductor **501b** is formed from a conductive material. The conductive material suitable for forming the first signal conductor **501a** and the second signal conductor **501b** may be a metal (e.g., a copper) or a metal alloy (e.g., a copper alloy). The first signal conductor **501a** and the second signal conductor **501b** may be formed in any suitable gauge, such as 28 AWG, 30 AWG or 32 AWG. The first signal conductor **501a** and the second signal conductor **501b** may be used to transmit differential signals. For example, the first signal conductor **501a** may be energized by a first voltage, and the second signal conductor **501b** may be energized by a second voltage that cooperates with the first voltage. The voltage difference between the first signal conductor **501a** and the second signal conductor **501b** represents a signal.

(26) Each of the first signal conductor **501a** and the second signal conductor **501b** is enclosed within a respective first insulator **503a** and second insulator **503b**. The first insulator **503a** and the second insulator **503b** may be bonded together, or separated from each other as shown. The first insulator **503a** and the second insulator **503b** run the entire length of the cable **500**, except for the portion that are removed at the end portion of the cable **500** in order to establish an electrical connection with the cable connector **1**. Each of the first insulator **503a** and the second insulator **503b** may be formed from an insulative material such as polypropylene (PP).

(27) A shield conductor **505** is disposed around the first signal conductor **501a**, the second signal conductor **501b**, the first insulator **503a** and the second insulator **503b**. In other words, the first signal conductor **501a**, the second signal conductor **501b**, the first insulator **503a** and the second insulator **503b** are enclosed within the shield conductor **505**. The shield conductor **505** may run the entire length of the cable **500**, except for the portion that are removed at the end portion of the cable **500** in order to establish an electrical connection with the cable connector **1**. That is, the shield conductor **505** may provide shielding for the first signal conductor **501a** and the second signal conductor **501b** over the entire length of the cable **500**, except for the portion that is removed at the end portion of the cable **500** in order to establish an electrical connection with the cable connector **1**. The shield conductor **505** may be formed of a conductive material, such as aluminized biaxially oriented PET film.

(28) An outer insulator **507** is disposed around the first signal conductor **501a**, the second signal conductor **501b**, the first insulator **503a**, the second insulator **503b** and the shield conductor **505**. In other words, the first signal conductor **501a**, the second signal conductor **501b**, the first insulator **503a**, the second insulator **503b** and the shield conductor **505** are enclosed within the outer insulator **507**. The outer insulator **507** may run the entire length of the cable **500**, except for the portion that are removed at the end portions of the cable **500** in order to establish an electrical connection with the cable connector **1**. The outer insulator **507** may be referred to as the “sheath” of the cable **500**. The outer insulator **507** may be formed of an insulative material, such as polyvinyl chloride (PVC).

(29) It should be appreciated that the cable **500** is only exemplary and may be in any suitable form

to provide the desired signal transmission performance. Although the first signal conductor **501a** and the second signal conductor **501b** are shown in FIGS. 2A and 2B as extending parallel to each other in the cable **500** without twisting around each other, it should be appreciated that the first signal conductor **501a** and the second signal conductor **501b** may twist around each other along a longitudinal direction of the cable **500** with a certain lay length. In the illustrated embodiment, cable connector **1** terminates **18** cables all configured as cable **500**. In some embodiments, one or more of the cables may have a different configuration. Also, it should be appreciated that FIG. 2A shows one end of a cable **500** prepared for termination to a cable connector **1**. A second end of the cable may be similarly prepared for termination or may be terminated in any other way, suitable for making connections to an electronic component within an electronic system.

(30) Referring back to FIGS. 1A and 1C, the terminal retention mechanism **300** is configured for retaining a plurality of conductive terminals **200** in multiple, here two, terminal rows **200a** and **200b** mutually opposed and spaced apart. It should be appreciated that the conductive terminals **200** of the cable connector **1** may be arranged in any other numbers of terminal rows, such as a single row or more than two rows.

(31) FIGS. 3A and 3B schematically illustrate a terminal row **200a** of the conductive terminals **200** of the cable connector **1**. As shown in FIGS. 3A and 3B, the conductive terminals in the terminal row **200a** are substantially aligned therein and include a plurality of pairs of signal terminals (which may also be referred to as “signal pins”) (“S”) and a plurality of ground terminals (which may also be referred to as “ground pins”) (“G”) **201**. Each of the signal terminals and the ground terminals **201** is formed of a conductive material. The conductive material suitable for forming the conductive terminals may be a metal (e.g., a copper) or a metal alloy (e.g., a copper alloy).

(32) Each pair of signal terminals of the plurality of pairs of signal terminals includes a first signal terminal **203a** and a second signal terminal **203b**. The first signal terminal **203a** and the second signal terminal **203b** may have the same configuration. The first signal terminal **203a** and the second signal terminal **203b** may form a differential signal pair for transmitting differential signals. For example, the first signal terminal **203a** may be energized by a first voltage, and the second signal terminal **203b** may be energized by a second voltage that cooperates with the first voltage. The voltage difference between the first signal terminal **203a** and the second signal terminal **203b** represents a signal.

(33) Each of the first signal terminal **203a**, the second signal terminal **203b** and the ground terminal **201** includes a contact portion **205**, a tail portion **207** and an intermediate portion, here shown as body portion **209**, extending between the contact portion **205** and the tail portion **207**. The tail portion **207** is configured for electrically connecting with the end portion of the cable **500**, and the contact portion **205** is configured for electrically connecting with a conductive portion of a mating component, such as a pad on a circuit board (not shown), which will be described in detail below. In the illustrated embodiment, the contact portions **205** are shaped as beams. The contact portions **205** curve into socket **103** such that they are deflected upon insertion of a mating component in socket **103** to generate force for making electrical and mechanical contact to pads on the mating component. A pair of signal terminals (i.e., the first signal terminal **203a** and the second signal terminal **203b**) is arranged between two adjacent ground terminals **201**. In this way, the ground terminals **201** are able to separate two adjacent pairs of signal terminals to reduce crosstalk between the two adjacent pairs of signal terminals, thereby improving signal integrity. These conductive terminals are aligned in the terminal row **200a** in a “G-S-S . . . G-S-S-G” pattern as shown in FIGS. 3A and 3B, with two adjacent pairs of signal terminals separated by a ground terminal.

(34) The tail portions **207** of the ground terminals **201** extend beyond the tail portions **207** of the first signal terminals **203a** and the second signal terminals **203b** and are connected together through the crossbar **211**. The tail portions **207** of the ground terminals **201**, the tail portions **207** of the first signal terminals **203a** and the second signal terminals **203b**, and the crossbar **211** are substantially

in the same plane. The tail portions **207** of the first signal terminal **203a** and the second signal terminal **203b** are disposed in a U-shaped space **213** defined by the tail portions **207** of two adjacent ground terminals **201** and the crossbar **211**. Connecting the tail portions **207** of the ground terminals **201** together through the crossbar **211** may provide a conductive or partially conductive pathway between the ground terminals **201**, thereby bridging the ground terminals **201** together. This may control or damp undesired resonances occurring within the ground terminals **201** during operation of the cable connector **1**, thereby improving signal integrity. The crossbar **211** may be formed from any suitable material. In some examples, the crossbar **211** may be formed from the same type of material that forms the ground terminals **201** or other suitable conductive material. In some other examples, the crossbar **211** may be formed of an electrically lossy material. For example, the crossbar **211** may be molded from or comprise an electrically lossy material. However, in the illustrated embodiment, crossbar **211** is of the same material and thickness as the ground terminals **201** and is stamped from the same sheet of metal in the same operation as ground terminals **201**.

(35) FIG. **4** schematically illustrates the terminal retention mechanism **300** of the cable connector **1**. As shown in FIG. **4**, the terminal retention mechanism **300** is an elongated T-shaped body and includes a vertically extending retention portion **301** and a boss portion (which may also be referred to as a “table portion”) **303** extending laterally from the retention portion **301**. The retention portion **301** is configured for being disposed around the plurality of conductive terminals **200** to retain the plurality of conductive terminals **200** in the two terminal rows **200a** and **200b** as described above. The terminal retention mechanism **300** may be formed of an insulative material. Examples of insulative materials that are suitable for forming the terminal retention mechanism **300** include, but are not limited to, plastic, nylon, liquid crystal polymer (LCP), polyphenylene sulfide (PPS), high temperature nylon or polyphenylene oxide (PPO) or polypropylene (PP).

(36) FIG. **4** illustrates the terminal retention mechanism **300** without terminals **200** shown. In some embodiments, terminal retention mechanism **300** may be molded as an insulative body and then terminals **200** may be inserted into openings in the insulative body. In other embodiments, terminal retention mechanism **300** may be formed as one or more insulative members molded around respective groups of terminals **200**. One insulative member, for example, may be molded around each row of terminals **200**.

(37) Regardless of how retention mechanism **300** is integrated with terminals **200**, table portion **303** may be shaped such that at least an upper surface of the tail portions **207** of the signal conductors is exposed for termination to conductive structures of cables. FIG. **5** schematically illustrates, in a perspective view, a plurality of the cables shown in FIG. **2A** being connected to the conductive terminals of the cable connector shown in FIG. **1A**, wherein the insulative housing **100** and the shield mechanism **400** of the cable connector **1** are removed.

(38) As shown in FIG. **5**, the body portions **209** (which are obscured in FIG. **5** by the retention portion **301**) of the conductive terminals in each of the two terminal rows **200a** and **200b** are retained in the retention portion **301**, and the tail portions **207** of the conductive terminals protrude from the retention portion **301** and rest on a corresponding one of a first side **305a** (FIG. **4**) and a second side **305b** (FIG. **4**) opposite to the first side **305a** of the boss portion **303**. That is, the boss portion **303** separates the tail portions **207** of the conductive terminals in the two terminal rows **200a** and **200b**. Each terminal row is arranged on a corresponding side of the boss portion **303** in such a way that the tail portions **207** of the ground terminals **201**, the tail portions **207** of the first signal terminals **203a** and the second signal terminals **203b** and the crossbar **211** are substantially in the same plane. The tail portions **207** of the first signal terminal **203a** and the second signal terminal **203b** are disposed in a U-shaped space **213** defined by the tail portions **207** of two adjacent ground terminals **201** and the crossbar **211**. In some examples, the terminal retention mechanism **300** may be molded directly around the conductive terminals. In some other examples, the retention portion **301** and the boss portion **303** may be formed separately and then assembled

together.

(39) With continuing reference to FIG. 5, a plurality of cables **500** (in the figure, eighteen cables) are connected to the plurality of conductive terminals **200** of cable connector **1**. In particular, each cable **500** is processed at an end portion in such manner that a pair of signal conductors (i.e., the first signal conductor **501a** and the second signal conductor **501b**) extend beyond the shield conductor **505**. Each signal conductor of the pair of signal conductors of each cable **500** is connected to the tail portion **207** of a corresponding one pair (i.e., the first signal terminal **203a** and the second signal terminal **203b**) of the plurality of pairs of signal terminals, thereby enabling transmission of differential signals between the cable **500** and the cable connector **1**. Each signal conductor may be connected to the tail portion **207** of a corresponding signal terminal by soldering. As shown in FIG. 5, solder **600** may be applied to the connection area where the signal conductor is connected to the tail portion **207** of the corresponding signal terminal and form a solder joint, so as to connect the signal conductor to the tail portion **207** of the corresponding signal terminal and establish an electrical connection therebetween. It should be appreciated that any other suitable method, such as welding, may be used to connect the signal conductor to the tail portion of the corresponding signal terminal. The shield conductor **505** is held on and in contact with the crossbar **211**, as will be described in detail below in conjunction with FIG. 1A.

(40) FIG. 6 schematically illustrates the shield mechanism **400** of the cable connector **1**. The shield mechanism **400** includes a first shield member **401** configured for mounting onto and in contact with the ground terminals **201** and/or crossbar **211** (e.g., by laser welding) and across the signal conductors (i.e., the first signal conductor **501a** and the second signal conductor **501b**) and the signal terminals (i.e., the first signal terminal **203a** and the second signal terminal **203b**), and holding the shield conductors **505** on the crossbar **211** such that the shield conductor **505** is in contact with the crossbar **211**. Holding the shield conductor **505** of the cable **500** on the crossbar **211** can prevent the signal conductors from moving relative to the signal terminals, so as to improve the reliability of the connection between the signal conductors and the signal terminals and thus the reliability of the cable connector **1**. The first shield member **401** may be formed of any suitable shielding material. As shown, the valley portions **405** may be attached to the ground terminals **201** and the signal conductors of the cable and the portions of the tails of the signal terminals of the connector may fit within the peak portions **403**.

(41) As shown in FIG. 6, the first shield member **401** is in the form of a pair of first shield plates **401a** and **401b**. Each of the pair of first shield plates **401a** and **401b** extends longitudinally at least along the entire length of the corresponding terminal row and laterally at least along the tail portions **207** of the plurality of ground terminals **201** of the corresponding terminal row and the crossbar **211**. When the cables **500** are connected to the plurality of conductive terminals **200** of the cable connector **1**, the pair of first shield plates **401a** and **401b** is mounted (e.g., by laser welding) from outside of the two terminal rows **200a** and **200b** onto and in contact with the plurality of ground terminals **201** of the corresponding terminal row, respectively, and press against and are in contact with the shield conductors **505**, thereby holding the shield conductors **505** on the crossbar **211** such that the shield conductors **505** are in contact with the crossbar **211**. As such, the first shield member **401**, the ground terminals **201**, the crossbar **211** and the shield conductors **505** together form a shield around the connection area where the signal conductors are connected to the signal terminals, thereby improving the signal integrity of the signal transmission between the cable **500** and the cable connector **1**.

(42) As shown in FIG. 6, each of the pair of first shield plates **401a** and **401b** is a wave-shaped plate having a peak portion **403** and a valley portion **405**. Each valley portion **405** is mounted (e.g., by laser welding) onto and in contact with a corresponding ground terminal **201** of the plurality of ground terminals **201**. The first shield plates **401a** and **401b** may overlap the crossbar **211** and in some embodiments may be welded to crossbar **211** (not shown in FIG. 6; see, e.g., FIG. 3A). Each peak portion **403** spans the signal conductors and the signal terminals and defines a chamber **407**.

The end portion of the cable **500** is received in the chamber **407**, and the peak portion **403** presses against and is in contact with the shield conductor **505** to hold the shield conductor **505** on the crossbar **211**. Each peak portion **403** is formed with an aperture **409** open to the chamber **407** to allow at least a connection area where the signal conductors are connected to the tail portions **207** of the corresponding signal terminals to be accessible via the aperture **409**. The aperture **409** allows for checking, via it, the integrity of the connection area where the signal conductors are connected to the tail portions of the corresponding signal terminals, for example checking the integrity of the solder joint. Furthermore, the aperture **409** also allows for checking, via it, whether there is a short circuit between this connection area and the first shield member **401**, for example checking whether there is a short circuit between the solder **600** and the first shield member **401**. This can improve the reliability of the cable connector **1**.

(43) As shown in FIGS. **1A** and **6**, a first portion of each peak portion **403** in contact with the shield conductor **505** is formed with a first retention mechanism **411** to improve a holding force on the shield conductor **505**. In some examples, as shown, the first retention mechanism **411** includes a protrusion protruding into the chamber **407** from the first portion. The protrusion may be an embossment formed by pressing the peak portion **403** toward the chamber **407** from the other side of the peak portion **403**. It should be appreciated that the first retention mechanism **411** may be a barb protruding from the first portion into the chamber **407**, or in any other suitable form. It should also be appreciated that a second portion of the crossbar **211** in contact with the shield conductor **505** may be formed with a second retention mechanism (not shown) to improve a holding force on the shield conductor **505**. Similar to the first retention mechanism **411**, the second retention mechanism may be a protrusion or barb protruding from the second portion into the chamber **407**, or in any other suitable form.

(44) As shown in FIGS. **4** and **5**, the retention portion **301** of the terminal retention mechanism **300** includes a first positioning protrusion **307** protruding onto the first side **305a** and the second side **305b** of the boss portion **303**. The first positioning protrusion **307** is configured for cooperating with the peak portion **403** to enable the pair of first shield plates **401a** and **401b** to be accurately positioned relative to the plurality of ground terminals **201** and preventing the pair of first shield plates **401a** and **401b** from moving in a direction along the first side **305a** and the second side **305b** relative to the plurality of ground terminals **201** when the pair of first shield plates is mounted onto the plurality of ground terminals **201**.

(45) As shown in FIG. **6**, the shield mechanism **400** may also include a second shield member **413** configured for being disposed between the two terminal rows **200a** and **200b** in the terminal retention mechanism **300**. The second shield member **413** may be formed of any suitable shielding material. The second shield member **413** extends longitudinally at least along the entire length of the two terminal rows **200a** and **200b** and laterally at least along the tail portions **207** of the plurality of ground terminals **201** and the crossbar **211**. The second shield member **413** can electrically isolate the two terminal rows **200a** and **200b** and form a shield between the two terminal rows **200a** and **200b**. This can reduce crosstalk between high speed signals in the two terminal rows **200a** and **200b**, thereby improving a signal integrity. This allows the two terminal rows **200a** and **200b** to be arranged at a small interval relative to each other, thereby reducing the overall size of the cable connector **1**. The second shield member **413** together with the first shield member **401**, the ground terminals **201**, the crossbar **211** and the shield conductors **505** form a shield around the connection area where the signal conductors are connected to the tail portions of the corresponding signal terminals, thereby further improving the signal integrity of the signal transmission between the cable **500** and the cable connector **1**.

(46) As shown in FIG. **6**, the second shield member **413** may be in the form of a perforated shield plate. In some examples, as shown in FIGS. **4** and **5**, the boss portion **303** of the terminal retention mechanism **300** may be formed with a slot **311** for receiving the second shield member **413**. In some other examples, the terminal retention mechanism **300** may be molded directly around the

second shield member **413** to hold the second shield member **413** therein. The second shield member **413** is connected to the crossbar **211** and/or the pair of first shield plates **401a** and **401b** (not shown), or is electrically floated by being insulated from the crossbar **211** and the pair of first shield plates **401a** and **401b**.

(47) Referring back to FIGS. **1A** to **1D**, the insulative housing **100** includes a first housing portion **100a** and a second housing portion **100b** configured to be mounted to the first housing portion **100a**. FIGS. **7A** and **7B** schematically illustrate the first housing portion **100a** and the second housing portion **100b** of the insulative housing **100**, respectively. The insulative housing **100** may be formed of an insulative material. Examples of insulative materials that are suitable for forming the insulative housing **100** include, but are not limited to, plastic, nylon, liquid crystal polymer (LCP), polyphenylene sulfide (PPS), high temperature nylon or polyphenylene oxide (PPO) or polypropylene (PP). It should be appreciated that the first housing portion **100a** and the second housing portion **100b** may be formed from the same insulative material or from different insulative materials.

(48) As shown in FIG. **7A**, the first housing portion **100a** defines a terminal cavity **101** therein. The retention portion **301** of the terminal retention mechanism **300** is configured to be received in the terminal cavity **101** of the first housing portion **100a**. The boss portion **303** of the terminal retention mechanism **300** is configured to protrude from the first housing portion **100a** and to be received in the second housing portion **100b** when the retention portion **301** is received in the terminal cavity **101**.

(49) The second housing portion **100b** is configured for holding the first shield member **401** (shown as the pair of first shield plates **401a** and **401b**) in position when being mounted to the first housing portion **100a**. This allows the first shield member **401** to be reliably mounted onto the ground terminals **201** and the shield conductors **505**. The second housing portion **100b** may be secured to the first housing portion **100a** by a dual injection molding. It should be appreciated that the second housing portion **100b** may also be formed and assembled with the first housing portion **100a** by other suitable means.

(50) In some examples, as shown in FIGS. **1A** and **1C**, when the retention portion **301** is received in the terminal cavity **101**, the contact portions **205** of the conductive terminals in each terminal row **200a** and **200b** extend cantilevered from the retention portion **301** opposite to the tail portions **207** and into the terminal cavity **101**, and together with the contact portions **205** of the conductive terminals in the other terminal row define therebetween a receiving space **S** for receiving the conductive portion of the circuit board. In this case, a first socket **103** is formed in the first housing portion **100a** and opens to the terminal cavity **101**. The first socket **103** is configured for allowing a portion on or near the edge of the circuit board, such as a daughter card, to be inserted therethrough into the receiving space **S**, such that the conductive portion of the circuit board is inserted between and in contact with the contact portions **205** of the conductive terminals in the two terminal rows **200a** and **200b**. It should be appreciated that the terminals **200** of the cable connector **1** may be arranged in any other numbers of terminal rows. It should also be appreciated that the first housing portion **100a** may have any other numbers of sockets, such as two or more sockets. Although the two terminal rows **200a** and **200b** are illustrated in FIGS. **1A** and **1C** as being spaced apart in such a way that the conductive terminals are aligned with each other along an arrangement direction, it should be appreciated that the two terminal rows **200a** and **200b** may be spaced apart in such a way that the conductive terminals are staggered with each other along the arrangement direction.

(51) It should be appreciated that, in some other examples, the contact portions **205** of the conductive terminals of each terminal row **200a** and **200b** may extend cantilevered from the retention portion **301** opposite to the tail portion **207** and to the exterior of the first housing portion **100a**, and extend opposite to the contact portions **205** of the conductive terminal in the other terminal row outside the first housing portion **100a**. This allows the contact portions **205** to be mounted (e.g., by surface mounting technique) onto the conductive portion of the circuit board.

Although the contact portions **205** are shown as extending substantially parallel to the tail portions **207**, it should be appreciated that the contact portions **205** may extend at any suitable angle (e.g., a right angle) relative to the tail portions **207**.

(52) The retention portion **301** of the terminal retention mechanism **300** may include a first attachment structure **313**, and the first housing portion **100a** may include a second attachment structure **113**. The first attachment structure **313** and the second attachment structure **113** are configured for cooperating with each other to secure the retention portion **301** in the terminal cavity **101**. As shown in FIG. 4, the first attachment structure **313** includes a barb protruding from the retention portion **301**, and as shown in FIG. 7A, the second attachment structure **113** includes a notch formed in the first housing portion **100a** for receiving the barb. It should be appreciated that the first attachment structure **313** and the second attachment structure **113** may also be in any other suitable form.

(53) The retention portion **301** of the terminal retention mechanism **300** may include a first positioning structure **315**, and the first housing portion **100a** may include a second positioning structure **115**. The first positioning structure **315** and the second positioning structure **115** are configured to cooperate with each other to enable the retention portion **301** to be accurately positioned in the terminal cavity **101** when the retention portion **301** is placed into the terminal cavity **101**. As shown in FIG. 4, the first positioning structure **315** includes first positioning slots recessed into the retention portion **301** from two longitudinally opposite end faces **316a** and **316b** of the retention portion **301**, respectively, and as shown in FIG. 7A, the second positioning structure **115** includes first positioning ribs protruding into the terminal cavity **101** from inner walls of the terminal cavity **101**. It should be appreciated that the first positioning structure **315** and the second positioning structure **115** may be in any other suitable form.

(54) The first housing portion **100a** may include a first mounting structure **117**, and the second housing portion **100b** may include a second mounting structure **118**. The first mounting structure **117** and the second mounting structure **118** are configured for cooperating with each other to secure the first housing portion **100a** and the second housing portion **100b** together. As shown in FIGS. 1A and 7A to 7B, the first mounting structure **117** includes two T-shaped protrusions mutually opposed formed across the terminal cavity **101** on the first housing portion **100a**, and the second mounting structure **118** includes two T-shaped recesses recessed into the second housing portion **100b** from two longitudinally opposite end faces **121a** and **121b** of the second housing portion **100b**. Each of the two T-shaped protrusions snaps into a corresponding one of the two T-shaped recesses. It should be appreciated that the first mounting structure **117** and the second mounting structure **118** may be in any other suitable form.

(55) The boss portion **303** of the terminal retention mechanism **300** may include a third positioning structure **319**, and the first housing portion **100a** may include a fourth positioning structure **119**. The third positioning structure **319** and the fourth positioning structure **119** are configured for cooperating with each other to enable the retention portion **301** to be accurately positioned in the terminal cavity **101** when the retention portion **301** is placed into the terminal cavity **101**. As shown in FIG. 4, the third positioning structure **319** includes a pair of third positioning ribs protruding from two longitudinally opposite end faces **320a** and **320b** of the boss portion **303**, respectively. The pair of third positioning ribs defines a second positioning slot **321** therebetween. As shown in FIG. 7A, the fourth positioning structure **119** includes a second positioning rib formed on the two T-shaped protrusions described above. The second positioning rib is received in the second positioning slot **321** between the pair of third positioning ribs. It should be appreciated that the third positioning structure **319** and the fourth positioning structure **119** may also be in any other suitable form.

(56) Although the cable **500** is described above as a component separate from the cable connector **1**, it should be appreciated that the cable **500** may be an inherent part of the cable connector **1**, and together with the cable connector **1** form an electrical connection system. The cable **500** may be

connected to other cables or electronic components at the end portion opposite to the cable connector **1**.

(57) The cable connector **1** and the electrical connection system according to the present application can eliminate the need for an intermediate circuit board, thereby saving the costs for manufacturing and assembly of the electronic system, while providing good shielding for the signal transmission, thereby providing good signal transmission performance.

(58) Cable termination designs as disclosed herein, however, may be used in connection with connectors configured to serve other functions. For example, a connector may be constructed with a shielding mechanism at the cable termination as described herein with different shaped contact portions of the terminals **200**. The contact portions may be shaped, for example, as blades or pads, for example. The housing for such a connector may have a housing with an insulative tongue with the contact portions exposed in opposing surfaces of the tongues. The tongue may be sized to be inserted into a socket of a mating connector, such as socket **103**. In such a configuration, the exposed contact portions of one connector may align with beam-shaped contact portions of the other connector, so as to mate.

(59) It should be noted that the terms “first” and “second” in the description, the claims and the drawings of the application are used to distinguish similar objects, and are not necessarily used to describe a specific sequence. It should be understood that numbers used in this way can be interchanged under appropriate circumstances such that the embodiments of the present disclosure described herein can be implemented in a sequence other than those illustrated or described herein. The present disclosure is not limited to the details of construction or the arrangements of components set forth in the foregoing description and/or the drawings. Various embodiments are provided solely for purposes of illustration, and the concepts described herein are capable of being practiced or carried out in other ways. Also, the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” or “involving,” and variations thereof herein, is meant to encompass the items listed thereafter (or equivalents thereof) and/or as additional items.

Claims

1. A cable connector comprising: an insulative housing; a plurality of conductive terminals held by the insulative housing in a row, each conductive terminal comprising a contact portion extending beyond the insulative housing, a tail portion extending out of the insulative housing, and an intermediate portion at least partially embedded in the insulative housing; a bar joining the tail portions of a subset of the plurality of conductive terminals; and a shield plate separated from the plurality of conductive terminals by the insulative housing.
2. The cable connector of claim 1, wherein: the shield plate comprises a plurality of openings.
3. The cable connector of claim 1, wherein: the shield plate is electrically insulated from the bar.
4. The cable connector of claim 1, wherein: the insulative housing comprises a socket; and the contact portions of the plurality of conductive terminals curve into the socket of the insulative housing.
5. The cable connector of claim 1, wherein: the insulative housing comprises a retention portion disposed around the plurality of conductive terminals to retain the plurality of conductive terminals in the row, and a boss portion comprising a plurality of projections; and the tail portions of the plurality of conductive terminals are disposed on the plurality of projections of the boss portion of the insulative housing.
6. The cable connector of claim 1, comprising: a wave-shaped shield member comprising alternating peak portions and valley portions, wherein each valley portion is mounted to a respective conductive terminal.
7. The cable connector of claim 6, wherein: each peak portion of the wave-shaped shield member

spans a respective pair of conductive terminals of the plurality of conductive terminals.

8. The cable connector of claim 7, wherein: each peak portion of the wave-shaped shield member comprises an opening sized and positioned to expose at least a portion of the tail portion of the respective pair of conductive terminals of the plurality of conductive terminals.

9. A cable assembly comprising: an insulative housing; a plurality of signal terminals held by the insulative housing in a row, each signal terminal comprising a contact portion extending beyond the insulative housing, and a tail portion extending out of the insulative housing; a ground member comprising a bar disposed adjacent the tail portions of the plurality of signal terminals, and a plurality of ground terminals extending from the bar and between respective ones of the plurality of signal terminals; a plurality of cables, each cable comprising a signal conductor and a shield conductor surrounding the signal conductor, the shield conductor disposed on a solid portion of a surface of the bar of the ground member, the signal conductor comprising an end extending out of the shield conductor and mounted on the tail portion of a respective one of the plurality of signal terminals; a wave-shaped shield member comprising alternating peak portions and valley portions, each valley portion mounted to a respective ground terminal, each peak portion abutting the shield conductor of a respective cable of the plurality of cables; and a shield member, wherein the shield member and the plurality of cables are disposed on opposite sides of the ground member.

10. The cable assembly of claim 9, wherein: the wave-shaped shield member extends beyond the shield conductor of the respective cable of the plurality of cables toward the contact portions of the plurality of signal terminals.

11. The cable assembly of claim 9, wherein: each peak portion of the wave-shaped shield member comprises an opening aligned with the end of the signal conductor of a respective cable of the plurality of cables.

12. The cable assembly of claim 11, wherein: each peak portion of the wave-shaped shield member comprises a protrusion or barb protruding toward the shield conductor of the respective cable of the plurality of cables.

13. The cable assembly of claim 9, wherein: the bar of the ground member comprises a protrusion or barb protruding toward the shield conductor of a respective cable of the plurality of cables.

14. The cable assembly of claim 9, wherein: the shield member is electrically insulated from the ground member.

15. A cable assembly comprising: an insulative housing; a plurality of conductive terminals held by the insulative housing in a row, each conductive terminal comprising a contact portion extending beyond the insulative housing, a tail portion extending out of the insulative housing, and an intermediate portion at least partially embedded in the insulative housing; a bar joining the tail portions of a portion of the plurality of conductive terminals; a plurality of cables disposed on a first side of the bar, each cable comprising a signal conductor having an end mounted on the tail portion of a respective one of the plurality of conductive terminals, and a shield conductor surrounding the signal conductor and contacting the bar; and a shield member disposed on a second side of the bar opposite to the first side of the bar.

16. The cable assembly of claim 15, wherein: the plurality of conductive terminals comprise a plurality of pairs of signal terminals and a plurality of ground terminals disposed between respective pairs of the plurality of pairs of signal terminals; and the bar joins the tail portions of the plurality of ground terminals.

17. The cable assembly of claim 16, further comprising: a wave-shaped shield member comprising alternating peak portions and valley portions, wherein: each valley portion is mounted to the tail portion of a respective ground terminal of the plurality of ground terminals; and each peak portion spans a respective pair of the plurality of pairs of signal terminals.

18. The cable assembly of claim 17, wherein: the shield member comprises a plurality of first openings.

19. The cable assembly of claim 18, wherein: the wave-shaped shield member comprises a plurality of second openings extending through respective valley portions.
